

Month 1 - Foundations of AI & ML

Topics:

- Introduction to AI & ML Concepts
- Python Fundamentals for Data Science (Variables, Data Types, Control Flow, Functions, Libraries like NumPy & Pandas)
- Basic Statistics (Mean, Median, Mode, Variance, Standard Deviation)
- Data Visualization (Matplotlib & Seaborn)

Project:

- **Sales Data Analysis:** Analyze a provided sales dataset to identify trends, patterns, and key performance indicators. Visualize findings.

Resources:

- [Python for Everybody](<https://www.py4e.com/lessons>)
- [NumPy Official Documentation](<https://numpy.org/doc/stable/>)
- [Pandas Official Documentation](<https://pandas.pydata.org/docs/>)
- [DataCamp - Introduction to Data Science](<https://www.datacamp.com/courses/introduction-to-data-science>)

Practice:

- [HackerRank Python Challenges](<https://www.hackerrank.com/domains/python>)
- [Kaggle Learn - Pandas](<https://www.kaggle.com/learn/pandas>)

Month 2 - Machine Learning Algorithms & Model Building

Topics:

- Supervised Learning Algorithms (Linear Regression, Logistic Regression, Decision Trees, K-Nearest Neighbors)
- Model Evaluation Metrics (Accuracy, Precision, Recall, F1-Score)
- Data Preprocessing (Feature Scaling, Encoding Categorical Variables)
- Introduction to Scikit-learn library

Project:

- **Customer Churn Prediction:** Build a binary classification model to predict customer churn using a public dataset.

Resources:

- [Scikit-learn User Guide](https://scikit-learn.org/stable/user_guide.html)
- [Coursera - Machine Learning by Andrew Ng (Stanford University)](<https://www.coursera.org/learn/machine-learning>) (Focus on Week 1-7 for foundational algorithms)
- [Kaggle Learn - Machine Learning](<https://www.kaggle.com/learn/intro-to-machine-learning>)

Practice:

- [Kaggle Datasets & Competitions](<https://www.kaggle.com/datasets>) (Start with beginner-friendly "Titanic" or "House Prices" competitions)
- [Towards Data Science Articles on Algorithms](<https://towardsdatascience.com/>) (Search for specific algorithm explanations)

Month 3 - Deep Learning Fundamentals & Deployment Basics

Topics:

- Introduction to Neural Networks
- Deep Learning Libraries (TensorFlow/Keras or PyTorch basics)
- Basic Neural Network Architectures (Feedforward Networks)
- Introduction to Model Deployment Concepts (e.g., Flask API)
- Resume Building & Interview Preparation for ML roles

Project:

- **Image Classification (MNIST):** Build and train a simple neural network to classify handwritten digits from the MNIST dataset. Deploy it as a basic web API.

Resources:

- [TensorFlow/Keras Official Documentation](<https://www.tensorflow.org/guide>)
- [PyTorch Official Documentation](<https://pytorch.org/docs/stable/index.html>)
- [DeepLearning.AI TensorFlow Specialization on Coursera](<https://www.coursera.org/specializations/tensorflow-in-practice>) (Focus on the first course for practical NN implementation)
- [Heroku Free Tier for Deployment](<https://www.heroku.com/>) (For hosting simple web apps)

Practice:

- [LeetCode - Algorithm Problems](<https://leetcode.com/problemset/all/>) (Focus on practicing coding skills relevant to ML roles)
- [Kaggle Learn - Deep Learning](<https://www.kaggle.com/learn/deep-learning>)
- [Build your own predictive model and deploy it on a free tier service like Heroku](<https://www.heroku.com/>)